

Software Requirements Specification (SRS)

AgroFamily
(Formerly AgroConnect Cameroon)

Version 1.0

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1 Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive description of the AgroFamily platform. It outlines the functional and non-functional requirements, use cases, and feasibility studies for the development of a digital marketplace tailored to the Cameroonian agricultural sector.

1.2 Scope

AgroFamily is a web-based platform designed to bridge the gap between farmers, buyers, and service providers in Cameroon. It facilitates the buying and selling of crops, animals, and tools, while also providing a space for service providers (e.g., transport, labor) to offer their expertise.

1.3 Intended Audience

This document is intended for stakeholders, project managers, developers, and quality assurance teams involved in the development and maintenance of the AgroFamily system.

2 Overall Description

2.1 System Context

AgroFamily operates as a centralized hub for the agricultural community. It replaces fragmented traditional markets with a structured digital environment, ensuring better price transparency and wider market reach for rural farmers.

2.2 Major Functionalities

The platform provides a unified interface for multiple agricultural stakeholders:

- **User Management:** Multi-role support for Farmers, Buyers, and Service Providers.
- **Marketplace:** Dynamic product listings with categorization and search capabilities.
- **Transaction Management:** Secure escrow-based ordering system.
- **Communication:** Integrated notification and messaging system.
- **Feedback:** Review and rating system for products and services.

3 Specific Requirements

3.1 Functional Requirements

The system must provide the following key features:

- **FR1: User Authentication:** The system shall allow users to register and log in securely using JWT and bcrypt encryption.
- **FR2: Product Management:** Farmers shall be able to create, update, and delete listings for crops, seeds, animals, and tools.
- **FR3: Marketplace Browsing:** Buyers shall be able to filter products by category, subcategory, and location.
- **FR4: Shopping Cart:** The system shall allow users to add multiple items to a cart and manage quantities before checkout.
- **FR5: Escrow Payment System:** The system shall hold funds in an escrow account until the buyer confirms receipt of the product.
- **FR6: Service Listings:** Service providers shall be able to list services such as transport, veterinary care, and labor.
- **FR7: Notifications:** The system shall send real-time notifications for order status changes, promotions, and messages.
- **FR8: Review System:** Users shall be able to provide ratings (1-5) and text reviews for purchased products.

3.2 Non-Functional Requirements

The following performance and quality standards are required:

- **Performance:** The system shall respond to user queries within 2 seconds under normal load conditions.
- **Security:** All sensitive data, including passwords, shall be encrypted. The system shall implement protection against SQL injection and XSS attacks.
- **Usability:** The interface shall be mobile-responsive and follow intuitive design principles to accommodate users with varying technical skills.
- **Reliability:** The system shall maintain 99.5% uptime during peak agricultural seasons.

3.3 Feasibility Studies

- **Economic Feasibility:** The project is economically viable as it leverages open-source technologies (MERN/PERN stack variants) and addresses a high-demand market in Cameroon with minimal overhead costs.
- **Security Feasibility:** Implementation of industry-standard authentication (JWT) and secure payment gateways (Escrow) ensures the platform's integrity and user trust.

4 Use Cases

The following table summarizes the primary user interactions with the system:

Use Case	Description
UC1: Farmer Lists Product	Farmer uploads product details, price, and images to the marketplace.
UC2: Buyer Purchases Item	Buyer adds items to cart, selects a payment method, and initiates an escrow transaction.
UC3: Service Provider Offers Service	Provider lists transport services available for farmers to book.
UC4: Admin Manages Categories	Admin adds or modifies product categories to keep the marketplace organized.

5 Appendices

5.1 Glossary

- **Escrow:** A financial arrangement where a third party holds funds before they are transferred to the seller.
- **JWT:** JSON Web Token, used for secure authentication.

5.2 References

- GitHub Repository: AgroConnect-Cameroon